

1 Immune gene and protein profiles across blood, cerebrospinal 2 fluid and peripheral nerve in inflammatory neuropathies

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12 **Abstract**

13 **Background**

14 Blood, cerebrospinal fluid and peripheral nerve capture different aspects of inflammatory
15 neuropathy. We examined immune gene and protein profiles in Guillain Barré syndrome and
16 chronic inflammatory demyelinating polyradiculoneuropathy using public transcriptomic
17 datasets and published proteomic and genetic findings. We compared gene modules across
18 three acute blood cohorts, pooled published cerebrospinal fluid protein coefficients from two
19 cohorts, and analysed nerve expression by donor. The nerve comparison included nine donors
20 with chronic inflammatory demyelinating polyradiculoneuropathy and 11 with chronic
21 idiopathic axonal polyneuropathy. Published genetic candidates were mapped to nerve cell
22 groups.

23 **Results**

24 CXCL8 recruitment and complement scores were higher in acute disease in all three blood
25 cohorts. Pooled Hedges g estimates were 1.08 (95% confidence interval -0.63 to 2.78) and
26 1.11 (-1.73 to 3.95), respectively. Nine of ten selected cerebrospinal fluid proteins had
27 positive coefficients in both cohorts and pooled q values below 0.05. CD5 differed in

direction between cohorts and had a pooled q value of 0.984. In nerve macrophages, the Fc receptor module score was lower in inflammatory demyelinating neuropathy than in the axonal disease comparator. Its q value was 0.0087 across eight macrophage modules and 0.087 across all 80 nerve module comparisons. The difference persisted after adjustment for age and centre and decreased after adjustment for macrophage composition. Four signalling and receptor genes contributed to the lower score. Correlations among nerve modules and with disability had q values above 0.05. *CDH4* expression was strongest in Schwann cell states, whereas *GNG7* expression was strongest in B cells.

Conclusions

These datasets show recurring recruitment and complement expression in acute blood, inflammatory protein changes in cerebrospinal fluid, and distinct cellular expression patterns in peripheral nerve. The macrophage Fc receptor finding depends on the scope of multiple testing correction and on cellular composition. Genetic candidate localization identifies nerve cell populations for further study.

Keywords

Inflammatory neuropathy, transcriptomics, proteomics, blood nerve barrier, macrophages, genetic susceptibility

Background

Guillain Barré syndrome (GBS) and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) differ in clinical course and diagnostic criteria [1,2,3]. Antibodies, complement and macrophages contribute to nerve injury, with their roles varying across diseases and subtypes [4,5,6,7,8,9]. Peripheral nerve atlases and experimental studies have also described diverse immune and Schwann cell responses to inflammation and injury [10,11,12,13]. Leukemia inhibitory factor (LIF), for example, influences Schwann cell

52 survival, proliferation and migration in experimental systems [14,15]. Together, these
53 findings provide a basis for examining immune and repair gene expression in inflammatory
54 neuropathies.

55 The blood nerve barrier (BNB) regulates exchange between the circulation and the
56 endoneurium. Endothelial junctions and associated mural cells help maintain its specialized
57 properties [16,17,18]. Experimental studies have implicated leukocyte integrins, endothelial
58 adhesion molecules and chemokines in immune cell trafficking across this interface
59 [19,20,21]. Changes in junctional protein expression have also been reported in CIDP nerve
60 [22]. The BNB therefore provides an anatomical setting in which circulating immune cells
61 can interact with the peripheral nerve environment.

62 Blood transcriptomic studies describe systemic immune responses in GBS using whole blood,
63 leukocytes and peripheral blood mononuclear cells (PBMCs) [23,24,25]. Cerebrospinal fluid
64 (CSF) proteomics provides information about inflammation and molecular injury, while
65 nerve transcriptomics identifies the cell populations expressing related genes [26,27]. Other
66 studies have reported monocyte changes in acute inflammatory demyelinating
67 polyradiculoneuropathy (AIDP), altered peripheral immune profiles in CIDP and associations
68 between blood expression and treatment response [28,29,30]. Experiments with patient
69 immunoglobulin further support a role for the BNB in inflammatory neuropathy [31]. Genetic
70 studies offer a complementary way to identify candidate genes [32]. Bringing these resources
71 together can help relate systemic immune expression to the cells and pathways represented in
72 peripheral nerve.

73 We compared immune gene modules across acute GBS blood cohorts, synthesized selected
74 published CSF protein findings, and examined immune and repair modules in CIDP nerve.
75 We then mapped published CIDP genetic candidates to nerve cell groups. Each dataset

76 contributed measurements from its own participants, allowing us to compare recurring
77 biological themes while accounting for differences in diagnosis, specimen and assay.

78 **Methods**

79 **Study design and datasets**

80 We analysed human data from blood, CSF, peripheral nerve and genetic studies. GSE211225
81 contained six whole blood profiles from acute GBS, ten from the postacute phase and six
82 from healthy controls [23]. GSE31014 contained leukocyte microarrays from seven patients
83 with GBS and seven healthy controls [24]. PRJNA1293757 contained PBMC transcriptomes
84 from three untreated patients with AIDP and two healthy controls [25]. We used the acute
85 disease and control groups from these three datasets to compare blood module effects.

86 The study associated with PXD002911 reported CSF protein and endogenous peptide
87 identifications by diagnostic group [26]. GSE285983 comprised 365,708 nuclei from 37 sural
88 nerves, including nine donors with CIDP and 11 with chronic idiopathic axonal
89 polyneuropathy (CIAP) [27]. The published human BNB transcriptome associated with
90 GSE107574 provided a reference for interpreting endothelial expression [17]. We selected
91 genetic candidates from the genome association, Mendelian randomization and colocalization
92 results reported by Du et al. [32]. Additional CSF reports described complement, quantitative
93 proteomics and inflammatory proteins [33,34,35]. We extracted adjusted protein coefficients
94 from the independent CSF discovery and replication cohorts reported by Kmezic et al. [36].
95 Published findings associated with PXD056286 supplied histological and proteomic
96 observations from 55 nerve biopsies in CIDP and its variants [37]. Table 1 summarizes the
97 resources and their analytical roles. Supplementary Data 1 lists database names, accession
98 identifiers, original filenames and source links.

99 **Gene modules**

100 Seven blood modules represented CXCL8 recruitment through CXCR1 and CXCR2, LIF and
101 oncostatin M (OSM) signalling through gp130, complement, Fc receptors, interferon
102 signalling through JAK and STAT, inflammatory monocytes and transendothelial migration.
103 Eight nerve modules represented BNB identity, leukocyte transmigration, CXCL8
104 recruitment, LIF signalling, complement, Fc receptors, macrophage state and Schwann cell
105 myelin repair. The analysis scripts specify the genes in each module. Module scores
106 summarize the expression of these gene sets. We assigned CSF findings to biological
107 categories using annotations in the source publications.

108 **Blood cohort analysis**

109 Within each blood cohort, we standardized gene expression across participants and averaged
110 the available genes in each module. Hedges g quantified differences between acute GBS and
111 healthy controls. Exact label permutation tests compared mean scores, and the Benjamini and
112 Hochberg procedure corrected P values across seven modules within each cohort [38]. We
113 combined effects from whole blood, leukocytes and monocyte pseudobulk using a model
114 with random effects. The DerSimonian and Laird estimator quantified variance between
115 studies. Confidence intervals used the modified Hartung and Knapp method, with the residual
116 variance scale bounded below by one and a t distribution with one fewer degree of freedom
117 than the number of cohorts. We reported τ^2 and I^2 to describe heterogeneity [39,40]. We
118 repeated the synthesis after omitting each cohort in turn. Postacute profiles contributed to the
119 original preprocessing, but the inferential comparisons concerned acute GBS and controls.
120 Donor independence between the acute and postacute phases could not be established from
121 the archived design.

122 **CSF protein synthesis**

123 We assembled CSF evidence from selected published reports. The quantitative synthesis
124 included all ten proteins other than IL8 that had coefficients and standard errors for both
125 cohorts in Table 4B of Kmezic et al. [36]. Eligibility depended on the availability of paired
126 estimates, irrespective of direction or significance. The source models adjusted for age, sex
127 and sample handling. We pooled coefficients using inverse variance weights in a model with
128 a fixed effect, calculated 95% confidence intervals using a normal approximation, and
129 applied the Benjamini and Hochberg procedure to the ten pooled tests. We also reported the
130 DerSimonian and Laird estimates, Q and I^2 to assess heterogeneity. IL8 was reported
131 separately in the source study and provided contextual evidence. Protein and peptide
132 identifications from PXD002911 and three additional reports formed a descriptive evidence
133 map [26,33,34,35]. Counts in this map indicate the number of selected studies and assay
134 platforms covering each biological category. The CSF analyses used published summary
135 results.

136 **Nerve pseudobulk analysis**

137 We grouped GSE285983 nuclei by donor and annotated cell state. The groups included
138 macrophages, T and natural killer cells, B cells, granulocytes, endoneurial endothelial cells,
139 other endothelial cells, pericytes, perineurial cells, endoneurial stroma, and Schwann cells in
140 myelinating, nonmyelinating, and repair and damage states. Groups with at least 20 nuclei per
141 donor were eligible. We summed counts within each donor and cell group, calculated counts
142 per million (CPM) using the total counts across all genes, and transformed expression as
143 $\log_2(\text{CPM} + 0.5)$. Donors were the biological replicates in comparisons between CIDP and
144 CIAP [41,42,43].

145 **Nerve modules and macrophage sensitivity analyses**

146 Within each cell group, we standardized each gene across eligible donors from all diagnostic
147 categories and averaged the available gene z scores to obtain module scores. We compared
148 CIDP with CIAP using the Mann and Whitney U test and expressed the difference as the
149 mean CIDP score minus the mean CIAP score. Multiple testing correction covered eight
150 modules within each cell group. We also applied correction across all 80 eligible nerve
151 module comparisons. Cell fractions were compared separately at donor level, with correction
152 across cell groups. These fractions describe the relative representation of each population in
153 the sampled nuclei [44].

154 The macrophage Fc receptor module contained 11 measured genes. *FCGR2C*, the twelfth
155 specified gene, was absent from the expression matrix. We assessed individual genes with
156 correction across 11 tests and four functional summaries with a separate correction.
157 Sensitivity analyses restricted the comparison to male donors, standardized genes within the
158 20 CIDP and CIAP donors, adjusted for age and centre, and added sex to the adjusted model.
159 Linear models used HC3 covariance estimates and residual degrees of freedom for
160 confidence intervals. An exact permutation test preserved the number of CIDP donors within
161 each centre. Supplementary Methods provides the model specifications.

162 We matched the original macrophage annotations to refined immune annotations using
163 nucleus barcodes. Composition summaries retained nuclei with unmatched labels and those
164 reassigned to other categories. We represented donor composition using centred log ratios
165 and added its first two principal components to the model containing age and centre. We also
166 reaggregated raw counts for all 20 CIDP and CIAP donors under alternative macrophage
167 definitions. Tests of refined subtypes required at least 20 nuclei per donor and at least three
168 eligible donors in each diagnosis group. Multiple testing correction covered the nine eligible
169 refined subtypes.

170 **Nerve module and disability correlations**

171 We paired five module scores within each of the nine CIDP donors. These were endothelial
172 LIF, endothelial leukocyte transmigration, macrophage complement, macrophage Fc
173 receptors, and myelin repair in Schwann cells annotated as repair and damage states.
174 Spearman correlations described their ten pairwise relationships. We also tested 15
175 combinations of cell group and module for associations with Inflammatory Neuropathy
176 Cause and Treatment (INCAT) disability scores. Multiple testing correction was applied
177 separately to the ten intermodule correlations and the 15 disability correlations.

178 **Comparison across compartments**

179 We compared recurring biological programs across compartments. A program entered the
180 evidence map if it had a consistent effect direction across blood cohorts, a corresponding
181 published CSF protein or peptide annotation, or expression in a relevant nerve cell
182 population. The map presents blood effects, pooled CSF coefficients, descriptive protein
183 findings and nerve expression as separate measurements. This approach compares biological
184 themes across independently sampled cohorts without calculating a single combined score.

185 **Genetic candidate localization**

186 We selected *CDH4*, *DIRAS1*, *GNG7* and *SLC39A3* from the genetic study of 516 CIDP cases
187 and 403,545 controls [32]. Candidate selection used the *CDH4* locus reported in women and
188 the published Mendelian randomization and colocalization findings for the other three genes.
189 We examined their expression across nerve cell groups and compared CIDP with CIAP.
190 CellBender counts from GSE285983 were normalized to 10,000 counts per nucleus,
191 transformed as $\log_2(\text{CP10k} + 1)$, and summarized by donor and cell group. We calculated
192 mean expression and the proportion of nuclei with detectable expression for each donor. For
193 each gene and cell group, 5,000 bootstrap resamples of CIDP donors provided intervals

194 around mean expression. Differences between CIDP and CIAP were tested using the Mann
195 and Whitney U test, with correction within each cell group. Genetic association estimates
196 were taken from the source publication, and the analyses performed here concerned cellular
197 expression.

198 **Statistical analysis**

199 Participants and nerve donors were the units of statistical inference. All hypothesis tests used
200 both tails. We denote P values adjusted by the Benjamini and Hochberg procedure as q and
201 use $q < 0.05$ within each stated testing family. The Fc component and sensitivity analyses
202 were planned after the initial macrophage association had been observed and are exploratory.
203 Scripts and software specifications are provided with the analysis archive. Supplementary
204 Methods describes the inputs and intermediate tables needed to reproduce each component.

205 **Writing and code support**

206 ChatGPT (OpenAI) assisted with manuscript editing, reference checking, formatting and
207 preparation of analysis and plotting code. The authors retain responsibility for the work.

208 **Results**

209 **Datasets and analytical roles**

210 The three blood cohorts provided estimates from whole blood, leukocytes and monocytes.
211 Published CSF studies contributed protein measurements, and the sural nerve atlas provided
212 expression profiles by donor and cell group. The BNB reference and genetic findings
213 supported the interpretation and selection of nerve genes (Fig. 1 and Table 1).

214 **Immune gene modules in acute GBS blood**

215 Acute GBS was associated with higher scores for six of the seven modules in GSE211225
216 after multiple testing correction. These were CXCL8 recruitment through CXCR1 and

217 CXCR2 (Hedges $g = 1.25$, $q = 0.045$), LIF and OSM signalling through gp130 ($g = 2.19$, $q =$
218 0.0076), complement ($g = 2.49$, $q = 0.0076$), Fc receptors ($g = 1.36$, $q = 0.038$), interferon
219 signalling through JAK and STAT ($g = 1.07$, $q = 0.045$), and inflammatory monocytes ($g =$
220 1.93 , $q = 0.015$). The transendothelial migration score was also higher ($g = 0.92$, $q = 0.126$).
221 Together, these results describe a broad increase in immune gene expression in acute GBS
222 whole blood.

223 Six of the seven module effects were positive in the independent leukocyte cohort
224 GSE31014. CXCL8 recruitment, complement, Fc receptors, transendothelial migration,
225 interferon signalling and inflammatory monocytes changed in the same direction as in whole
226 blood. In PRJNA1293757 monocytes, CXCL8 recruitment had the largest positive effect ($g =$
227 1.97), followed by a positive complement effect. The remaining module effects varied in
228 direction.

229 CXCL8 recruitment and complement had positive effects in all three cohorts. The pooled
230 Hedges g was 1.08 for CXCL8 recruitment (95% confidence interval -0.63 to 2.78 , $P =$
231 0.114 , $I^2 = 0\%$) and 1.11 for complement (95% confidence interval -1.73 to 3.95 , $P = 0.234$,
232 $I^2 = 57.2\%$). All seven pooled confidence intervals included zero. Four other modules had
233 positive effects in two cohorts, while LIF and OSM signalling was positive only in whole
234 blood (Fig. 2 and Table 2).

235 Omitting each cohort in turn left the pooled direction positive for CXCL8 recruitment ($g =$
236 0.97 to 1.41) and complement ($g = 0.48$ to 1.55). Both individual effects in each retained pair
237 were positive, although the intervals based on two cohorts were wide. Every other module
238 had at least one cohort with a discordant effect direction.

239 Heterogeneity differed across modules. I^2 was 0% for CXCL8 recruitment and interferon
240 signalling, 1.1% for transendothelial migration, 49.5% for inflammatory monocytes, 57.2%

241 for complement, 72.7% for Fc receptors and 81.2% for LIF and OSM signalling. The largest
242 variation therefore concerned pathways with strong effects in whole blood and smaller or
243 opposite effects in the other blood populations. With three cohorts, platform, cell composition
244 and clinical differences remain overlapping explanations for this variation.

245 **Published CSF protein findings**

246 Nine of the ten selected CSF proteins had positive coefficients in both cohorts. They were
247 SELE, IL2RA, CCL3, CR2, CD1C, THBD, NRP1, CD38 and IL6. Each had a pooled q value
248 below 0.05 after correction across ten proteins. CD5 had a discovery coefficient of -0.0882
249 and a replication coefficient of 1.51, giving a pooled coefficient of 0.0024 (95% confidence
250 interval -0.237 to 0.242 , $q = 0.984$). Heterogeneity was greatest for IL2RA ($I^2 = 94.9\%$),
251 CD5 (89.0%), IL6 (80.7%) and THBD (79.5%). In the source study, SELE and IL2RA were
252 the two proteins in this subset that met the adjusted significance threshold in both cohorts
253 [36]. Panels A and B of Figure 3 present the individual and pooled coefficients.

254 Four additional CSF reports described complementary protein findings. The study associated
255 with PXD002911 contributed protein and peptide identifications and functional annotations
256 [26]. D'Aguanno et al. described complement and inflammatory proteins [33], Ding et al.
257 reported proteins related to protein handling and neural structure [34], and Sun et al. reported
258 inflammatory proteins alongside routine CSF measurements [35]. Local inflammation was
259 represented in three reports and protein accumulation in two. Structural proteolysis,
260 complement, and protein handling and neural structure were each represented in one report
261 (Fig. 3C).

262 **Immune and repair gene expression in CIDP nerve**

263 The sural nerve atlas localized immune and repair genes to distinct cell populations.
264 Macrophages expressed *C3*, *C3AR1* and *FCGR2A*. Endoneurial endothelial cells expressed

265 *CLDN5*, *ICAM1*, *VCAM1*, *LIFR*, *IL6ST* and *OSMR*. Schwann cell states expressed gp130
 266 receptor components together with genes related to myelin, stress and repair (Fig. 4A).
 267 Endothelial *CLDN5* and adhesion gene expression was consistent with the cellular context
 268 described in the published human BNB transcriptome [17].

269 The largest statistical difference between CIDP and CIAP concerned the macrophage Fc
 270 receptor module. Its mean score was lower in CIDP by 0.673 z score units ($P = 0.0011$, $q =$
 271 0.0087 across eight macrophage modules). Macrophage leukocyte transmigration and
 272 macrophage state scores were also lower, with differences of -0.345 ($P = 0.0185$, $q = 0.074$)
 273 and -0.368 ($P = 0.0334$, $q = 0.089$), respectively. In perineurial cells, the macrophage state
 274 and complement scores were higher by 0.758 and 0.348, respectively, with $q = 0.109$ for
 275 both. The endothelial BNB identity score was lower by 0.404 ($P = 0.0276$, $q = 0.221$).
 276 Correction across all 80 eligible nerve module comparisons gave $q = 0.087$ for the
 277 macrophage Fc receptor result (Fig. 4B).

278 All nine CIDP donors were male, compared with seven of the 11 CIAP donors. The Fc
 279 receptor score remained lower in the comparison restricted to male donors (difference
 280 -0.687 , $P = 0.0007$). Adjustment for age and centre gave a difference of -0.690 (95%
 281 confidence interval -1.040 to -0.339 , $P = 0.0008$). The exact permutation test within centres
 282 gave $P = 0.0048$. Estimates also remained negative after individual donors, centres or genes
 283 were omitted in turn (Supplementary Fig. 3).

284 Four of the 11 measured genes had lower expression with $q < 0.05$. These were *FCGR2A*,
 285 *FCER1G*, *TYROBP* and *HCK*. *FCGRT* had $q = 0.713$, while *FCGR1A* had a positive estimate
 286 with $q = 0.148$. The summary of signalling components was lower ($q = 0.039$). The activating
 287 receptor, inhibitory receptor and neonatal Fc receptor summaries had q values above 0.05.
 288 The lower composite score thus reflected differences among particular receptor and
 289 signalling genes.

290 Adding two macrophage composition components to the model containing age and centre
291 reduced the estimated difference to -0.310 (95% confidence interval -1.038 to 0.419 , $P =$
292 0.375). The refined macrophage union also had a lower score ($P = 0.0062$). All nine eligible
293 individual refined subtype tests and all 32 annotation fraction tests had q values above 0.05
294 (Supplementary Fig. 4).

295 Cell fraction comparisons across the broad nerve groups all had q values above 0.05. The
296 largest numerical increases in CIDP were in nonmyelinating Schwann cells (0.064),
297 endoneurial stroma (0.027) and myelinating Schwann cells (0.013), with $q = 0.515$ for each
298 (Fig. 4C).

299 Among the nine CIDP donors, the strongest association with INCAT disability concerned the
300 myelin repair score in Schwann cells annotated as repair and damage states (Spearman $\rho =$
301 0.840 , $P = 0.0046$, $q = 0.069$). Endothelial LIF scores had a weaker positive association ($\rho =$
302 0.602 , $P = 0.086$). All 15 disability correlations had q values above 0.05 (Fig. 4D).

303 Published tissue findings provided further context for the nerve expression results. Stascheit
304 et al. reported terminal complement complex deposition on endoneurial capillaries in 52 of 55
305 biopsies from patients with CIDP or its variants [37]. This count corresponds to 94.5%, with
306 a calculated Wilson 95% confidence interval of 85.1% to 98.1%. The source study also
307 described macrophages expressing CD68, T cells expressing CD8, and increased abundance
308 of C3 and C6. Supplementary Figure 2 summarizes these published observations.

309 **Relationships among nerve modules**

310 Within the nine CIDP donors, endothelial LIF scores correlated positively with endothelial
311 leukocyte transmigration scores ($\rho = 0.667$, $P = 0.0499$, $q = 0.166$) and Schwann cell myelin
312 repair scores ($\rho = 0.783$, $P = 0.0125$, $q = 0.125$). Endothelial transmigration scores also

313 correlated with macrophage complement scores ($p = 0.683$, $P = 0.0424$, $q = 0.166$). All ten
314 intermodule correlations had q values above 0.05.

315 The evidence map brings these nerve findings together with the blood and CSF results.
316 CXCL8 recruitment and complement had consistent blood effect directions and
317 corresponding inflammatory or complement annotations in CSF. The Fc receptor module
318 score was higher in acute GBS whole blood than in healthy controls and lower in CIDP nerve
319 macrophages than in CIAP macrophages. Endothelial adhesion and Schwann cell repair
320 genes localized related programs within nerve, while CSF studies described protein
321 accumulation and proteolysis (Fig. 5 and Table 3).

322 **Cellular expression of genetic candidates**

323 The four genes selected from Du et al. represented two forms of published genetic evidence
324 [32]. *CDH4* was linked to the locus reported in women at 20q13.33. *DIRAS1*, *GNG7* and
325 *SLC39A3* were selected from the Mendelian randomization and colocalization findings.
326 Figure 5C displays the source estimates used for this selection.

327 In CIDP donor summaries, *CDH4* expression was highest in nonmyelinating Schwann cells
328 and Schwann cells in repair and damage states. *GNG7* expression was highest in B cells and
329 was also detected in macrophages and BNB endothelial cells. *SLC39A3* expression was low,
330 with its highest mean in BNB endothelium, while *DIRAS1* was sparsely expressed. Bootstrap
331 intervals describe variation among donors for each cell group (Supplementary Fig. 1). All
332 displayed comparisons between CIDP and CIAP had q values above 0.05 (Fig. 5B and D).

333 **Discussion**

334 This analysis describes immune gene and protein profiles in blood, CSF and peripheral nerve
335 using complementary public resources. CXCL8 recruitment and complement were the most
336 consistent blood modules in acute GBS. Selected CSF proteins showed concordant increases

337 across two cohorts. In nerve, immune and repair genes were distributed across macrophages,
338 endothelium and Schwann cells, and the macrophage Fc receptor score was lower in CIDP
339 than in CIAP. Published genetic candidates showed distinct cellular expression patterns.
340 Together, these findings connect recurring immune themes with the nerve populations in
341 which relevant genes are expressed.

342 The blood results suggest that recruitment and complement expression recur across different
343 blood preparations in acute GBS. Six modules were significant within the whole blood
344 cohort, whereas CXCL8 recruitment and complement retained positive directions in all three
345 cohorts and after successive cohort omissions. Their pooled confidence intervals were wide.
346 This pattern is compatible with the importance of chemokines and leukocyte trafficking
347 described in experimental studies and with published CSF findings concerning IL8
348 [19,36,45,46,47]. Blood module scores and CSF protein concentrations nevertheless represent
349 different measurements. Their agreement is most useful for selecting pathways to examine in
350 paired specimens.

351 The CSF findings provide protein evidence for inflammation and related vascular processes.
352 Nine of the ten selected proteins had positive coefficients in both cohorts, with SELE and
353 IL2RA also meeting the source study's adjusted significance threshold in each cohort [36].
354 The large heterogeneity for several proteins shows that the estimated magnitude varied
355 substantially between cohorts. The complementary proteomic and peptidomic reports
356 described inflammation, complement and molecular injury [26,33,34,35]. These findings
357 broaden the biological context of the transcriptomic results, while retaining the distinction
358 between the pooled analysis of ten reported proteins and the wider literature on CSF changes.

359 The anatomical setting also matters when interpreting these profiles. CSF samples the
360 environment surrounding nerve roots and the neuraxis, whereas the BNB is formed by
361 microvascular endothelial cells within peripheral nerve. Endothelial adhesion genes, immune

362 cell expression and Schwann cell repair programs place the nerve findings within a plausible
363 sequence of recruitment and tissue response. Because the specimens came from separate
364 cohorts, the timing of these events remains an open question. Paired blood and CSF
365 measurements, spatial nerve profiling and endothelial permeability assays could establish
366 how they relate within individual patients.

367 The macrophage Fc receptor result was the clearest nerve module association within the
368 original cell group analysis. Its direction persisted after demographic adjustment and
369 omission analyses, and four receptor or signalling genes contributed to the lower score. The
370 smaller estimate after composition adjustment indicates that the result depends partly on how
371 macrophage populations are represented in the model. This is relevant in light of the diverse
372 macrophage states described in the source atlas [27]. Published complement deposition on
373 endoneurial capillaries and peripheral B cell and CD8 effector changes in CIDP add further
374 evidence of immune involvement at different sites [29,37]. The lower Fc score relative to
375 CIAP should therefore be interpreted as a feature of this particular macrophage comparison.

376 The positive association between Schwann cell myelin repair scores and disability suggests a
377 relationship between the tissue response to injury and clinical impairment. Greater repair
378 gene expression could accompany more extensive damage or a greater demand for repair.
379 The q value of 0.069 and the small number of donors make its magnitude uncertain.

380 Correlations with endothelial modules offer related questions for studies that combine
381 transcript expression, tissue organization and clinical follow up.

382 Genetic candidate localization adds another perspective to the nerve atlas. The concentration
383 of *CDH4* expression in Schwann cell states and *GNG7* expression in B cells helps identify
384 cell populations in which these candidates can be studied. The *CDH4* association originated
385 from an analysis of women [32], whereas the CIDP nerve donors were all male. This
386 difference makes the cellular findings relevant to expression context while leaving sex

387 dependent genetic effects to be resolved. Low expression of *DIRAS1* and *SLC39A3* also
388 limits precise localization. Testing the effects of these genes in appropriate cell populations is
389 the next step in relating the published associations to nerve biology.

390 **Limitations**

391 The study has several limitations. The blood cohorts were small and differed in cell
392 composition and assay platform, producing wide pooled intervals. The CSF synthesis covered
393 ten proteins selected for reporting in the source publication, and the broader evidence map
394 was assembled from four selected studies. Its coverage is therefore restricted to those reports.
395 The pooled CSF tests used a normal approximation, which differs from the regression tests
396 used in the original small cohorts.

397 The nerve comparison included nine CIDP donors and 11 donors with an axonal neuropathy.
398 It therefore describes differences between two disease groups. All CIDP donors were male,
399 recruitment centres were imbalanced, and treatment or other clinical differences could
400 influence expression. The Fc receptor association had $q = 0.087$ across all 80 nerve module
401 comparisons and became smaller after composition adjustment. The component and
402 sensitivity analyses followed the initial observation, and subtype estimates were based on few
403 eligible donors. These features call for evaluation in a larger independent nerve cohort.

404 Blood, CSF and nerve came from separate cohorts, with diagnosis partly linked to specimen
405 type. The design thus compares disease and tissue profiles together. The companion spatial
406 panel contained none of the specified Fc receptor module genes, leaving that association
407 without spatial confirmation. Genetic estimates came from the published study, while our
408 analyses measured cellular expression. Mechanistic relationships among the compartments
409 remain to be tested with matched samples and functional experiments.

410 **Conclusions**

411 Public blood, CSF and nerve datasets reveal complementary immune expression patterns in
412 inflammatory neuropathies. CXCL8 recruitment and complement recur across acute GBS
413 blood cohorts, while CSF protein findings describe inflammation and related vascular
414 changes. The nerve atlas places immune and repair genes in specific cell populations and
415 identifies lower macrophage Fc receptor expression in CIDP than in CIAP. The estimate
416 varies with cellular composition, and its statistical support depends on the scope of multiple
417 testing correction. Localization of published genetic candidates adds a cellular basis for
418 further investigation. These results identify specific pathways and nerve populations for
419 studies using paired specimens and independent cohorts.

420 **List of abbreviations**

421 AIDP, acute inflammatory demyelinating polyradiculoneuropathy. BNB, blood nerve barrier.
422 CIAP, chronic idiopathic axonal polyneuropathy. CIDP, chronic inflammatory demyelinating
423 polyradiculoneuropathy. CPM, counts per million. CP10k, counts per 10,000. CSF,
424 cerebrospinal fluid. FDR, false discovery rate. GBS, Guillain Barré syndrome. GWAS,
425 genome wide association study. INCAT, Inflammatory Neuropathy Cause and Treatment.
426 LIF, leukemia inhibitory factor. OSM, oncostatin M. PBMC, peripheral blood mononuclear
427 cell.

428 **Declarations**

429 **Ethics approval and consent to participate**

430 This study analysed publicly available deidentified data and published results. The source
431 studies reported their ethics approvals and consent procedures. No new participants were
432 recruited and no new specimens were collected.

433 **Consent for publication**

434 Not applicable. The manuscript contains no identifiable personal information.

435 **Availability of data and materials**

436 Database names, accession identifiers, original filenames and source links are listed in

437 Supplementary Data 1, together with aggregate data underlying the figures and tables.

438 Supplementary Methods describes the analysis procedures and reconstruction of intermediate

439 tables. Acquisition dates and checksums are provided where they were recorded.

440 Analysis code and aggregate results are available in the GitHub repository and in the version

441 2.3.0 archive on Zenodo [48,49]. This release contains the CSF coefficient synthesis,

442 macrophage sensitivity analyses, annotation comparisons and source documentation used in

443 this manuscript. Complete recalculation requires the original inputs. The genetic localization

444 scripts additionally require the donor expression intermediate described in Supplementary

445 Methods.

446 **Competing interests**

447 The authors declare that they have no competing interests.

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452 **Authors' contributions**

453 YF conceived the study, developed the analytical framework, performed the analyses and

454 drafted the manuscript. FF curated data, validated results and revised the manuscript. SXM

455 contributed to validation, interpretation and manuscript revision. SMF contributed to data

456 curation, literature evaluation and manuscript revision. CJ supervised the methodology and
457 critically revised the manuscript. YH supervised the study, coordinated the project and
458 critically revised the manuscript.

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461 **Supplementary information**

462 Additional file 1. Supplementary_Data_1_v41.xlsx. Excel workbook containing the
463 aggregate source tables, sensitivity results and database access documentation cited as
464 Supplementary Data 1.

465 Additional file 2. Supplementary_Methods_v41.docx. Detailed analytical procedures and
466 reconstruction instructions, cited as Supplementary Methods.

467 Additional file 3. Supplementary_Figures_v41.docx. Supplementary Figures 1 to 4 with their
468 legends, showing candidate gene localization, published nerve findings and macrophage
469 sensitivity analyses.

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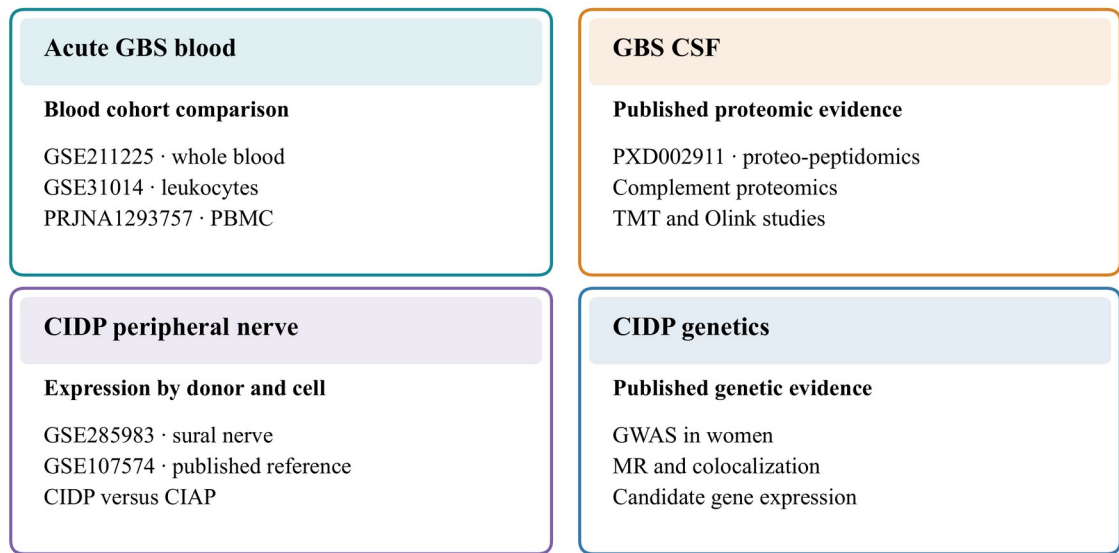
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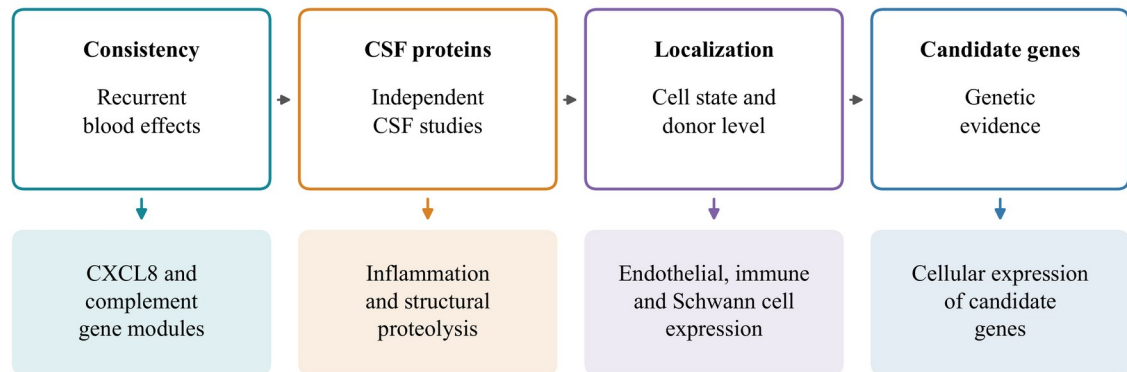
620 **Figures and legends**

621 **Figure 1. Study design and analytical roles**

A Data sources and analytical roles



B Complementary analyses



622

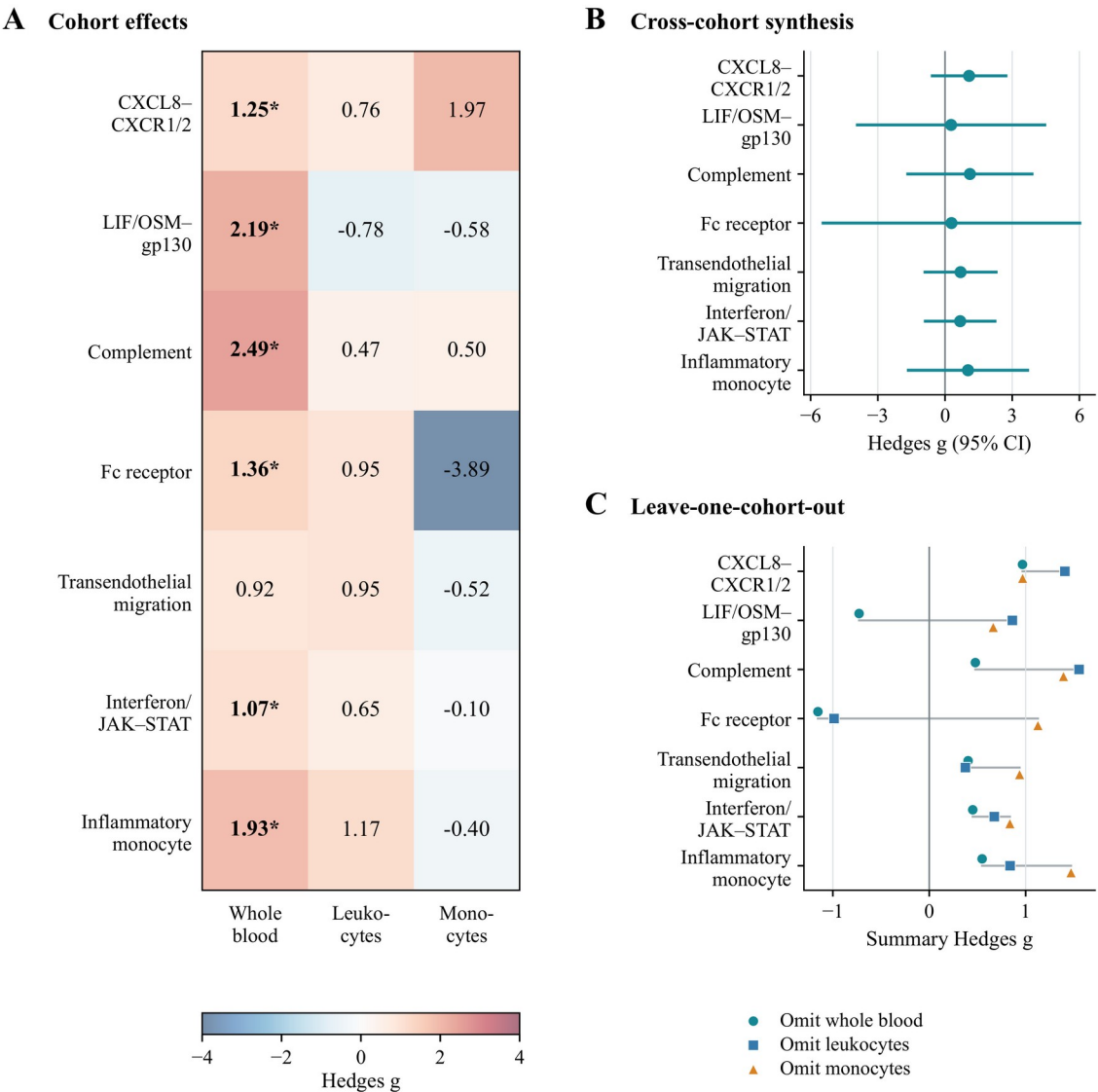
623 Public blood, CSF, peripheral nerve and genetic resources contributed complementary measurements.

624 Blood cohorts provided gene module effects, CSF reports provided protein and peptide findings, and the

625 nerve atlas provided expression by donor and cell group. Published genetic associations guided candidate

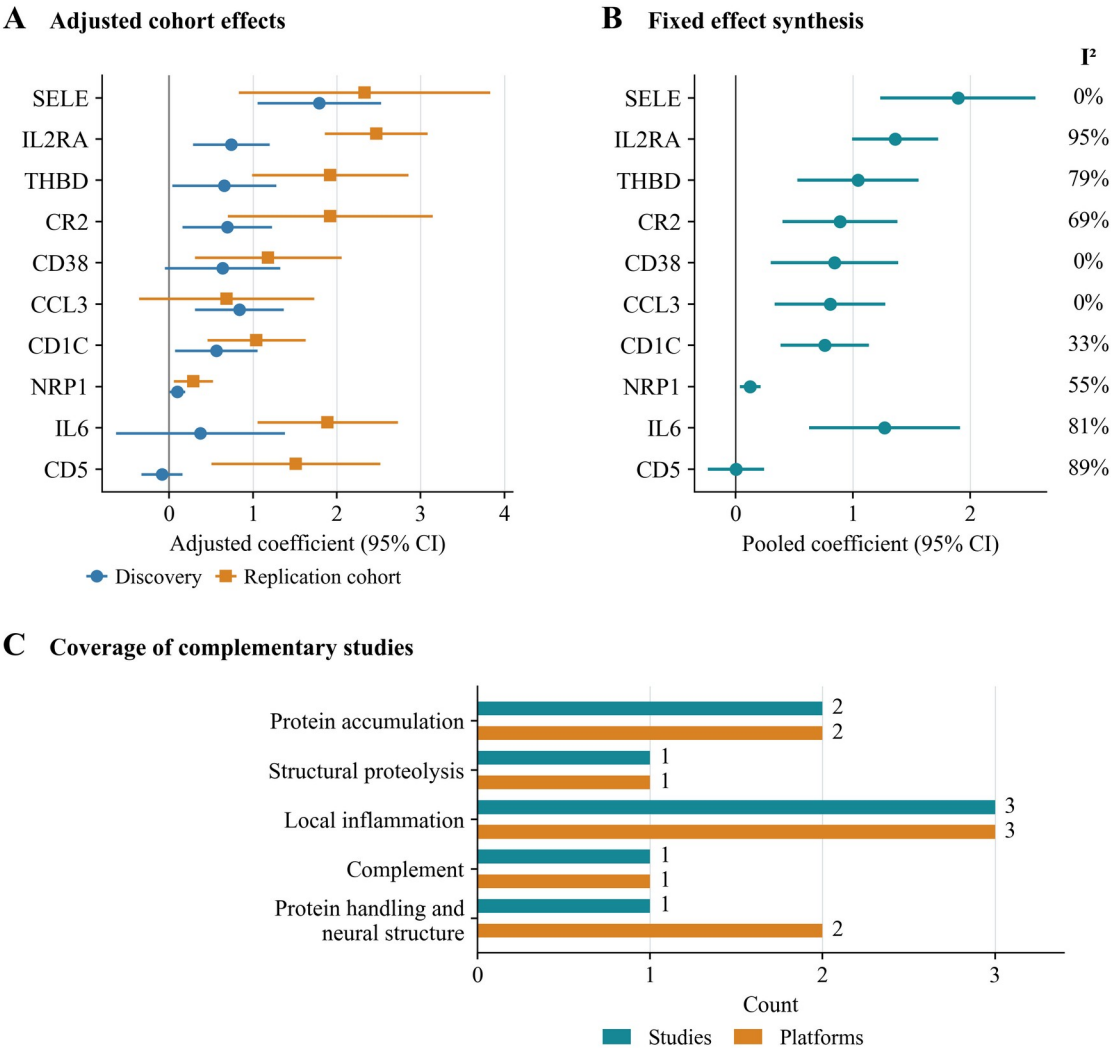
626 selection for nerve expression analysis. Arrows summarize analytical connections and biological questions.

627 **Figure 2. Acute GBS blood modules across cohorts**



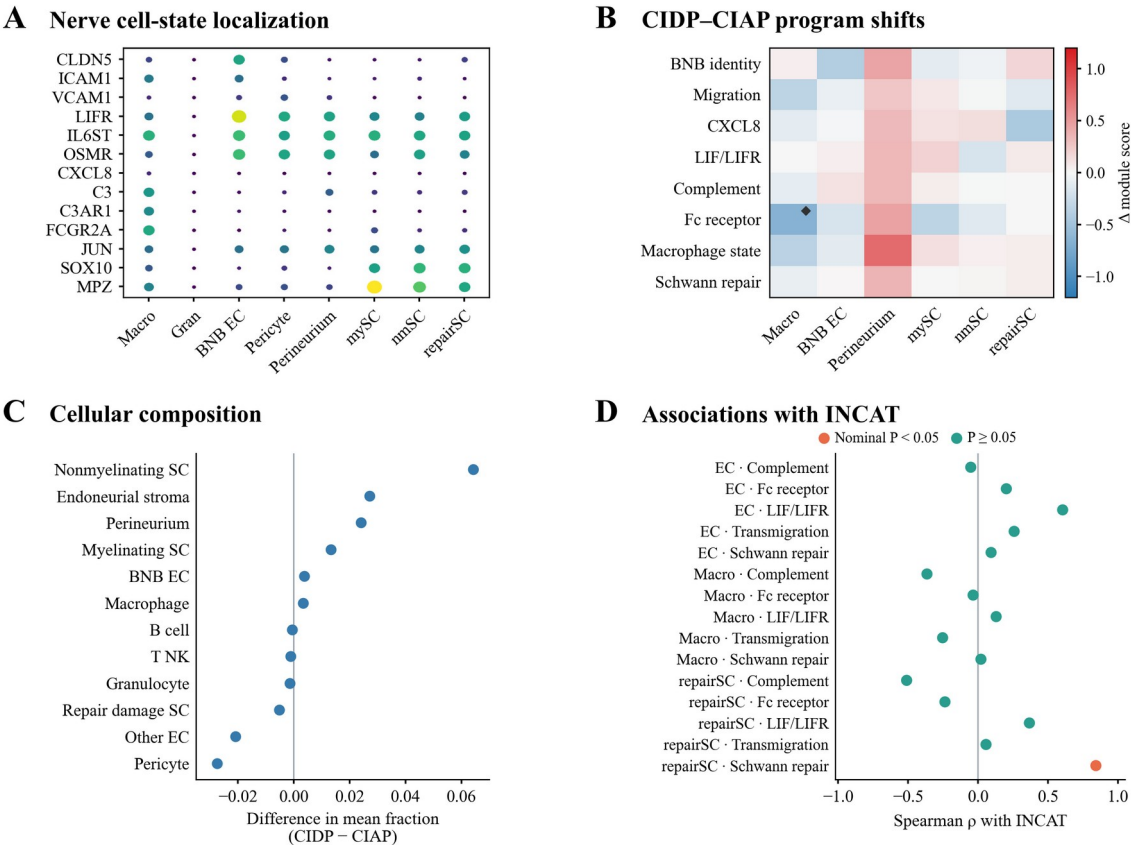
628
629 (A) Hedges g for seven gene modules in whole blood, leukocytes and monocyte pseudobulk. Asterisks
630 indicate $q < 0.05$ after correction within each cohort. (B) Pooled estimates from models with random
631 effects and 95% confidence intervals calculated using the modified Hartung and Knapp method. (C)
632 Pooled estimates after each cohort was omitted in turn. CXCL8 recruitment and complement retained
633 positive estimates in all subsets, and both effects in each retained pair were positive. All pooled intervals in
634 B include zero.

635 **Figure 3. Synthesis of published CSF protein coefficients**



636
637 (A) Coefficients from the discovery and replication cohorts of Kmezic et al., adjusted for age, sex and
638 sample handling [36]. Error bars show 95% confidence intervals calculated from the reported standard
639 errors using a normal approximation. (B) Estimates pooled with inverse variance weights in a model with a
640 fixed effect. The analysis includes all ten proteins other than IL8 with paired coefficients in source Table
641 4B. Nine pooled tests have $q < 0.05$, while CD5 has $q = 0.984$. I^2 describes heterogeneity. The source study
642 found adjusted significance in both cohorts for SELE and IL2RA. (C) Numbers of selected studies and
643 assay platforms covering each biological category, based on four complementary reports [26,33,34,35].
644 These counts describe the coverage of the evidence map.

645 **Figure 4. Immune and repair modules in CIDP nerve**



646

647 (A) Gene expression across nerve cell states. Dot size indicates the proportion of nuclei with detected

648 expression, and colour indicates scaled mean expression. (B) Differences in donor module scores between

649 CIDP and CIAP. Diamonds indicate $q < 0.05$ after correction within a cell group. The macrophage Fc

650 receptor result has $q = 0.087$ after correction across all 80 eligible comparisons. (C) Differences in mean

651 cell fractions, calculated as CIDP minus CIAP. The comparisons in B and C use the Mann and Whitney U

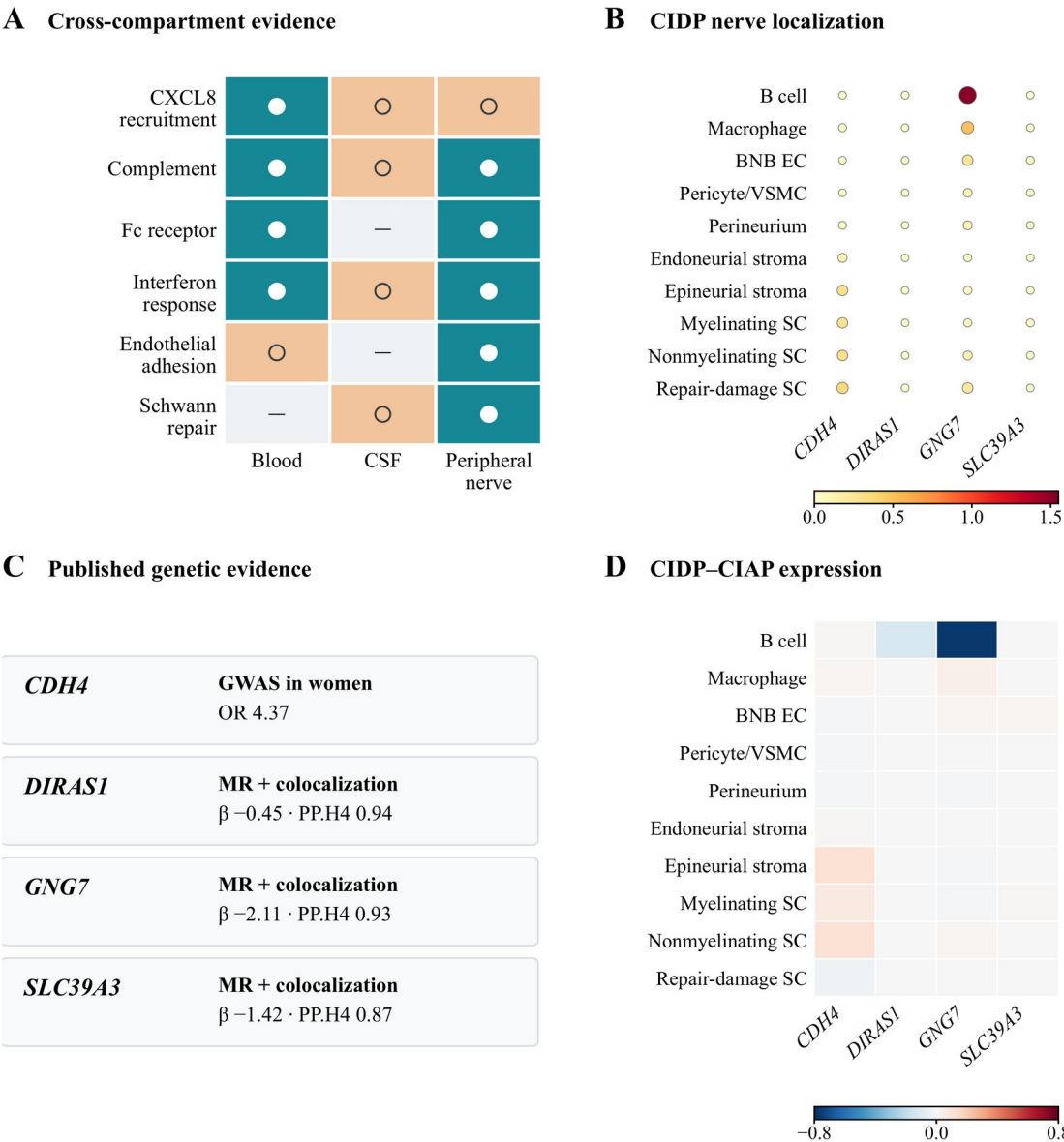
652 test with donors as independent samples. (D) Spearman correlations between donor module scores and

653 INCAT disability. Orange indicates $P < 0.05$ and teal indicates $P \geq 0.05$. All disability correlations have q

654 > 0.05 after correction. BNB EC denotes endoneurial microvascular endothelial cells. Supplementary

655 Figures 3 and 4 present the macrophage sensitivity analyses.

656 **Figure 5. Evidence across compartments and expression of genetic candidates**



657

658 (A) Descriptive evidence map. Filled white circles denote significance within an individual blood cohort or
659 the original nerve cell group testing family, or direct nerve expression localization. Open circles denote
660 recurring directions or representation in published findings. Horizontal marks indicate programs that were
661 not evaluated. The symbols summarize different forms of evidence. Blood pooled effects have $P > 0.05$,
662 and the nerve macrophage Fc receptor result has $q = 0.087$ across all 80 module comparisons. (B)
663 Expression of *CDH4*, *DIRAS1*, *GNG7* and *SLC39A3* across CIDP nerve cell groups. Dot size indicates the
664 percentage of nuclei expressing a gene, and colour indicates mean $\log_2(\text{CP10k} + 1)$, summarized by donor.
665 (C) Published genetic estimates from Du et al. that guided candidate selection [32]. (D) Differences in
666 donor mean expression between CIDP and CIAP. Red indicates higher expression in CIDP and blue
667 indicates lower expression. All displayed comparisons have $q > 0.05$. The colour scales show mean
668 expression in B and differences between diagnoses in D.

669 **Table 1. Datasets and analytical roles**

Resource	Material	Participants or samples	Analytical role
GSE211225	Whole blood transcriptomics	6 acute GBS, 10 postacute, 6 controls	Primary blood effects for individual participants
GSE31014	Leukocyte microarray	7 GBS, 7 controls	Independent assessment of blood effect direction
PRJNA1293757	PBMC transcriptomics	3 AIDP, 2 controls	Monocyte localization and assessment of effect direction
PXD002911	CSF proteomics and peptidomics	GBS (AIDP), MS and controls, with pooled discovery material	Published CSF identifications and functional annotations
GSE285983	Sural nerve nuclear transcriptomics	9 CIDP, 11 CIAP in focused comparison	Donor gene module scores, cell fractions and expression localization
GSE107574	Human endoneurial endothelial transcriptomics	Cultured cells and microvessels	Published BNB expression reference
Kmezic et al. 2023	CSF Olink proteomics	Discovery cohort with 6 GBS and 20 controls. Replication cohort with 6 GBS and 6 controls.	Synthesis of ten paired CSF protein coefficients from source Table 4B
PXD056286	Sural nerve histology and proteomics	55 biopsies from CIDP or its variants	Published tissue observations and calculated Wilson interval

670 GBS, Guillain Barré syndrome. AIDP, acute inflammatory demyelinating polyradiculoneuropathy. CIDP,
671 chronic inflammatory demyelinating polyradiculoneuropathy. CIAP, chronic idiopathic axonal
672 polyneuropathy. CSF, cerebrospinal fluid. MS, multiple sclerosis. PBMC, peripheral blood mononuclear
673 cell. BNB, blood nerve barrier. Eligible nerve donor numbers vary by cell group.

674 **Table 2. Pooled effects of acute GBS blood modules**

Gene module	Summary Hedges g	95% CI lower	95% CI upper	Modified Hartung and Knapp P	I ² , %	Cohorts, n
CXCL8 signaling via CXCR1 and CXCR2	1.075	−0.634	2.785	0.114	0.0	3
LIF and OSM signaling via gp130	0.271	−3.978	4.519	0.810	81.2	3
Complement	1.111	−1.728	3.950	0.234	57.2	3
Fc receptor	0.288	−5.510	6.086	0.851	72.7	3
Transendothelial migration	0.694	−0.958	2.347	0.212	1.1	3
Interferon signaling via JAK and STAT	0.677	−0.943	2.297	0.214	0.0	3
Inflammatory monocyte	1.026	−1.702	3.754	0.247	49.5	3

675 Positive Hedges g indicates a higher score in acute GBS. Variance between cohorts was estimated by the
676 DerSimonian and Laird method. Confidence intervals and P values use the modified Hartung and Knapp
677 procedure, with residual scale bounded below by one. Each estimate combines three cohorts. CI,
678 confidence interval.

679 **Table 3. Evidence across blood, CSF and peripheral nerve**

Program	Blood evidence	CSF evidence	Peripheral nerve evidence	Interpretation
CXCL8 recruitment	Positive in 3 cohorts. Pooled $P = 0.114$. whole blood $q < 0.05$	Published innate immune and inflammatory protein findings	Myeloid localization and endothelial trafficking context	Recurring recruitment expression
Complement	Positive in 3 cohorts. Pooled $P = 0.234$. whole blood $q < 0.01$	Published complement protein findings	Macrophage localization	Complement gene and protein findings
Fc receptor	Whole blood $q < 0.05$, with mixed directions across cohorts	Not directly evaluated	Macrophage $q = 0.0087$ within eight modules and $q = 0.087$ across 80 nerve comparisons	Lower macrophage score, attenuated after composition adjustment
Interferon response	Whole blood $q < 0.05$, with mixed directions across cohorts	Published evidence of antiviral defense	Macrophage, endothelial and Schwann cell states	Expression across different sampled populations
Endothelial adhesion	Positive blood direction	Not directly evaluated	ICAM1 and VCAM1 localization to BNB endothelial cells	Endothelial expression of adhesion genes
Schwann repair	Not evaluated	Published evidence of axonal domain peptide degradation	Repair and stress programs localized to Schwann cells	Structural injury and Schwann cell repair context

680 Blood comparisons concern acute GBS and healthy controls. The nerve comparison concerns CIDP and
681 CIAP. q denotes the adjusted P value within the stated testing family. The evidence map describes findings
682 from independently sampled cohorts.